
MA 527: Field and Galois Theory

Course Code	MA 527
Course Name	Field and Galois Theory
L-T-P-C	3-1-0-4
Prerequisites	MA 549 (Abstract Algebra)
Intended for	M.Sc./M.S./PhD/B.Tech
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **Module 1:** Fields, Characteristics and prime subfields, Field extensions, Automorphisms, Normal extensions and Splitting fields, Separable and Inseparable extensions, Algebraic closures (12 Hours)
- **Module 2:** Galois groups, The fundamental theorem of Galois theory, Finite fields, Cyclotomic extensions, Composite extensions, Norm and Traces, Cyclic extensions, Hilbert theorem 90 and Group cohomology, Kummer extensions (16 Hours)
- **Module 3:** Discriminants, Polynomials of degree 3 and 4, Ruler and Compass constructions, Solvability by radicals, Polynomials with Galois group S_n , Transcendental extensions, Solution of a cubic by Cardan's method, Solution of biquadratic by Ferrari's method (14 Hours)

Text Books

1. P. Morandi, **Field and Galois Theory**, Springer-Verlag, 1996.
2. D.S. Dummit and R. M. Foote, **Abstract Algebra**, 2nd Edition, John Wiley, 2002.

References

1. M. Artin, **Algebra**, Prentice Hall of India, 1994.
2. N. Jacobson, **Basic Algebra I**, 2nd Edition, Hindustan Publishing Co., 1984, W.H. Freeman, 1985.
3. S. Lang, **Algebra**, 3rd Edition, Springer (India), 2004.
4. J.S. Milne, Online notes: <https://www.jmilne.org/math/CourseNotes/FT.pdf>